

Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. The compressed paper-facing diagnostic has two theory-aligned readouts: ACPC tail risk (ATR), the high-quantile same-state clean/noisy action-conditioned rollout disagreement, and selective-margin pass rate (SMPR), a task-grounded near-boundary check that action-relevant distinctions remain separated. Across three independent LeWM training seeds and four control tasks, no-noise checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian robustness plateaus. At recovered endpoints, ATR drops sharply and SMPR rises to near one across tasks. The claim is a controlled diagnostic for matched-Gaussian robustness plateaus, not a closed-loop stability guarantee, universal perturbation-transfer result, or new training objective.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: <https://github.com/Anguo-star/lewm-acpc-diagnostics>

1 Introduction

1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 29]. For control, however, “irrelevant” is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is an incomplete target: planning uses the model’s *predictions*, so robustness must be stated after action-conditioned prediction.

We use this failure as a controlled diagnostic probe. Across three independent LeWM [19] training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher under observation-only Gaussian noise at std 0.08 with the goal unperturbed. Observation+goal noise is reported as a harder auxiliary stress condition. Test-time visual fragility is not new [15, 33, 12, 21]; our focus is what it reveals about JEPA latent world-model control.

1.2 Operationalizing action-conditioned predictive consistency

We operationalize a downstream-control view of visual robustness for JEPA world models through *action-conditioned predictive dynamics*. Let $z = E(h)$ and $\tilde{z} = E(\tilde{h})$ be the latents an encoder E produces from an unperturbed history h and a perturbed history $\tilde{h}$ of the same underlying state, and let $F^k(\cdot, \mathbf{a}_{t:t+k-1})$ be the action-conditioned latent predictor rolled out for k steps under an action sequence $\mathbf{a}$. We *allow* $z \neq \tilde{z}$. The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k(z, \mathbf{a}_{t:t+k-1}) \approx F^k(\tilde{z}, \mathbf{a}_{t:t+k-1}), \quad k = 1, \dots, H, \quad (1)$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. The requirement is *selective*: it should hold only for nuisance perturbations of the same underlying state. State

differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This diagnostic view changes what counts as a useful robustness measurement. Encoder geometry is a useful first-stage diagnostic because it shows whether visual perturbations enter and reshape the latent neighborhood, but encoder-level latent closeness, $\|z - \tilde{z}\| \rightarrow 0$, is not a complete control-robustness definition: large nuisance shifts can wash out after prediction, whereas small shifts can still be amplified downstream. Planning, cost, and action metrics remain useful as *downstream readouts* of predictive consistency; future objectives should regularize action-conditioned predictive dynamics while preserving action-relevant discriminability.

1.3 Existing noisy-evaluation results as a controlled probe

We use input-side noise augmentation as the controlled intervention. The $\sigma_{\max} \in \{0.01, \dots, 0.08\}$ sweep across four tasks creates paired baseline/recovered checkpoints for asking whether closed-loop Gaussian recovery is accompanied by lower same-state predictive tail risk and preserved task-grounded separations. Because the sweep forms broad task-dependent plateaus under finite evaluation variance, we do not treat a single point-best $\sigma_{\max}$ or small within-plateau score differences as meaningful checkpoint ordering.

1.4 Contributions

C1 — Selective ACPC principle. Robustness diagnostics should require both low same-state action-conditioned predictive-disagreement tails and preserved task-grounded near-boundary separations. Low consistency loss alone is not enough, because collapse can also make noisy and clean views agree.

C2 — Theory-matched compression. ATR estimates the ACPC-tail term exposed by the sampled-pool analysis and local Gaussian sensitivity calculation. SMPR estimates the discriminability/no-collapse countercondition that makes low ATR meaningful. The clean-margin term remains a formal caveat explaining why the diagnostic is not a closed-loop guarantee.

C3 — Fixed-checkpoint evidence. Across three LeWM training seeds, matched-Gaussian robustness plateaus coincide with large ATR reductions and near-one SMPR at recovered endpoints. Unseen-stressor scores bound the perturbation scope without turning the diagnostic into a method or universal transfer claim.

2 Related Work

2.1 Latent prediction and JEPA world models

JEPA [16] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 20] extend this idea to images, video, and dense physical-world representations. LeWM [19], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [6]. PLDM [25, 26], via `stable-worldmodel` [18], provides related latent-dynamics planning context. LeJEPA theory [14] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [31, 27, 35]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [29] and Littwin et al. [17] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [8] addresses a related invariance–equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [15, 33], SODA/DMC-GB [12], DreamerV3 [10], TD-MPC2 [11], and ViGMO [21] define future method-comparison axes; noisy/video JEPA variants [22, 13, 23] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [7], DBC [34], bisimulation-regularized MPC [24], and Bisim-JEPA [28] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [9, 30] and VIBR [5] likewise emphasize downstream control consequences. Wang et al. [32] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [4] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while task-grounded different-state pairs should remain separable. Section 3.2 then maps these two requirements to ATR and SMPR.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation-action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length- k prefix of this sequence after re-indexing time at t , and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \quad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \quad (2)$$

Let Π denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\text{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^H \alpha_k d(\Pi(\hat{z}_{t+k}), \Pi(\hat{\tilde{z}}_{t+k})), \quad \alpha_k \geq 0. \quad (3)$$

For the theoretical statements below, we use the induced horizon metric

$$d_H((u_1, \dots, u_H), (v_1, \dots, v_H)) = \sum_{k=1}^H \alpha_k d(u_k, v_k), \quad (4)$$

so that Equation (3) is exactly the clean/perturbed rollout-readout distance under the shared action sequence. In experiments, ATR uses the 90th percentile of this normalized same-state rollout disagreement. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder closeness is a useful first-stage risk signal, but it is neither necessary (a large but task-irrelevant encoder shift can wash out through F_θ and Π) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

Action-relevant discriminability (countercondition). Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future readout diverges,

$$\Delta_{\text{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y(Y^H(s_i, \mathbf{a}), Y^H(s_j, \mathbf{a})) > m, \quad (5)$$

the representation and predictions should keep them separable,

$$d(\Psi(z_i), \Psi(z_j)) > m', \quad (6)$$

where Ψ is a discriminability readout, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (3)–(6) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\text{ACPC}_H \rightarrow 0$ by collapsing the latent would violate (6), so the two conditions are inseparable.

3.2 Operational consistency diagnostics

The paper-facing diagnostic uses two readouts. ACPC Tail Risk (ATR) is the 90th percentile of normalized same-state clean/noisy ACPC-H rollout disagreement under a fixed recorded action sequence; lower is better. Selective Margin Pass Rate (SMPR) is the pass rate of task-grounded label-crossing pairs whose clean different-state rollout distance exceeds the same-state noisy radius at margin zero; higher is better. ATR measures predictive-stability tail risk, while SMPR rules out the collapse failure in which all states become indistinguishable. Encoder geometry and planner-side readouts are not separate paper-facing diagnostics in this compressed version.

Table 1: Theory-to-metric mapping for compressed selective ACPC.

Core theorem quantity	Paper-facing readout	Role
$\Pr[D > \epsilon] / \text{ACPC tail}$	ATR	same-state action-conditioned predictive-stability tail
$d_{\text{diff}} > d_{\text{same}} + m$	SMPR	task-grounded anti-collapse margin

3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g , define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \dots, \Pi(\hat{z}_{t+H}), g), \quad (7)$$

where J is the planner’s cost readout on the rollout-readout sequence. $C_{\tilde{h}}(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Proposition 1 (Encoder shift as one route to rollout disagreement). *Fix an action sequence $\mathbf{a}$ and write*

$$G_{\mathbf{a}}(z) = \Pi(F_{\theta}^{1:H}(z, \mathbf{a}))$$

for the rollout-readout map. If $G_{\mathbf{a}}$ is locally L_G -Lipschitz between $E_{\theta}(h)$ and $E_{\theta}(\tilde{h})$ under encoder metric d_Z and rollout-readout metric d_H , then

$$d_H(G_{\mathbf{a}}(E_{\theta}(h)), G_{\mathbf{a}}(E_{\theta}(\tilde{h}))) \leq L_G d_Z(E_{\theta}(h), E_{\theta}(\tilde{h})).$$

Proof. This is the local Lipschitz condition applied to the two encoded histories. $\square$

This bound makes encoder geometry a first-stage risk signal: small encoder shifts imply small rollout disagreement only under local insensitivity, large shifts can be contracted by $G_{\mathbf{a}}$, and small shifts can be amplified by the predictor or cost readout. This is why the empirical diagnostic measures the tail of action-conditioned rollout disagreement rather than encoder distance alone.

Proposition 2 (ACPC controls candidate-cost drift). *Assume that J is locally L_J -Lipschitz on the clean/perturbed rollout-readout neighborhood evaluated by the fixed candidate set under metric d_H . If*

$$d_H(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

Proof. The inequality is the local Lipschitz condition applied directly to the clean and perturbed rollout-readout sequences. $\square$

In the compressed diagnostic, this condition is instantiated by the L2 rollout-token distance whose high-quantile tail defines ATR; empirical companion audits are kept outside the paper-facing metric set.

Proposition 3 (Candidate top-1 stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j ,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. $\square$

Corollary 1 (ACPC-margin condition for candidate stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be a shared candidate set and assume the cost readout J is locally L_J -Lipschitz as in Proposition 2. If every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ between the clean and perturbed branches, and the clean branch has top-1/top-2 margin $\Delta > 2L_J\epsilon$, then the two branches select the same top-1 candidate from $\mathcal{A}$.*

Proof. Proposition 2 gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J\epsilon$ for every candidate. Applying Proposition 3 with $\eta = L_J\epsilon$ gives the result. $\square$

The corollary gives a formal link between ACPC and fixed-candidate decisions without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. The result is limited to a fixed candidate set; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

3.4 Sampled-pool stability and Gaussian sensitivity

The fixed-candidate statement is the deterministic core. A sampled candidate pool adds finite-sample risk. The next bound keeps the same diagnostic scope while exposing why ATR must be a tail readout and why clean candidate margins remain a caveat.

Theorem 1 (Sampled-pool ACPC stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool, and use deterministic tie-breaking for $\arg \min$. For each sampled candidate define*

$$D_j = d_H(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}^j)), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}^j))).$$

Assume J is locally L_J -Lipschitz on the evaluated rollout-readout neighborhood and that a single draw from q satisfies

$$\Pr_{\mathbf{a} \sim q}[D(\mathbf{a}) > \epsilon] \leq \delta.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean branch top-1/top-2 margin inside the sampled pool. Then

$$\Pr_{\mathcal{A}} \left[\arg \min_j C_h(\mathbf{a}^j, g) \neq \arg \min_j C_{\tilde{h}}(\mathbf{a}^j, g) \right] \leq K\delta + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon].$$

Theorem 1 is not a convergence theorem for adaptive planning. It controls the instability of one sampled candidate pool. The first term is the probability that at least one candidate has a large clean/noisy rollout discrepancy, which motivates ATR. The second term is the probability that the clean candidate pool has too small a top-1 margin; it explains why ATR is not a closed-loop guarantee. SMPR is separate from this flip bound: it is the selective-discriminability condition required to make low ATR meaningful rather than collapsed.

Corollary 2 (Replanning union bound). *If replanning step t has sampled-pool flip-probability bound*

$$r_t = K_t \delta_t + \Pr[\Delta_{\mathcal{A}_t} \leq 2L_J \epsilon_t],$$

then over T replanning steps,

$$\Pr[\exists t \leq T : \text{the sampled-pool top-1 flips at step } t] \leq \sum_{t=1}^T r_t.$$

Proposition 4 (Finite-sample tail calibration for paired diagnostics). *Fix the diagnostic sampling distribution used to draw independent paired clean/noisy histories and candidate action sequences. Let*

$$X_i = \mathbf{1}[D_i > \epsilon], \quad i = 1, \dots, n,$$

where D_i is the paired ACPC rollout-readout discrepancy for diagnostic item i . Let $p_\epsilon = \Pr[D > \epsilon]$ and $\hat{p}_\epsilon = n^{-1} \sum_i X_i$. Then, with probability at least $1 - \delta$ over the diagnostic sample,

$$p_\epsilon \leq \hat{p}_\epsilon + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

The same one-sided calibration applies to SMPR after fixing its task-grounded pair construction.

This calibration is with respect to the fixed diagnostic sampling distribution and its independent-sample idealization. It does not convert an empirical quantile or pass-rate table into a uniform bound over all closed-loop states; when nearby dataset windows are dependent, we use it only as calibration.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 5 (Local Gaussian ACPC sensitivity). *If E_θ and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_\theta(o)$ with bounded second-order remainders, then for small σ ,*

$$G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o)) J_E(o) \xi + R_\xi,$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_\xi \left[\|G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o))\|_2^2 \right] = \sigma^2 \|J_{G_{\mathbf{a}}}(E_\theta(o)) J_E(o)\|_F^2 + O(\sigma^3).$$

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Small encoder shifts can still be amplified by $J_{G_{\mathbf{a}}}$, while large nuisance shifts can become harmless after rollout if the product $J_{G_{\mathbf{a}}} J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while rollout readouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m],$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic calibration rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate-margin caveats, and collapse of task-grounded separations. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, or environment-feedback stability.

Selective ACPC pseudo-metric. For a nonempty fixed candidate family $\mathcal{A}$, the stability condition can be summarized by

$$D_{\text{ACPC}}^{\mathcal{A}}(h, \tilde{h}) = \max_{\mathbf{a} \in \mathcal{A}} d_H(\Pi(F_{\theta}^{1:H}(E_{\theta}(h), \mathbf{a})), \Pi(F_{\theta}^{1:H}(E_{\theta}(\tilde{h}), \mathbf{a}))). \quad (8)$$

When d_H is a metric (or a pseudo-metric, if some horizon weights vanish), $D_{\text{ACPC}}^{\mathcal{A}}$ is a pseudo-metric on histories after passing through the encoder, predictor, and readout: distinct histories can have zero distance if all candidate rollouts are readout-equivalent. This follows because each map $h \mapsto \Pi(F_{\theta}^{1:H}(E_{\theta}(h), \mathbf{a}))$ pulls back d_H to a pseudo-metric, and the pointwise maximum of pseudo-metrics is again a pseudo-metric. The useful object is therefore *selective* predictive stability. Small $D_{\text{ACPC}}^{\mathcal{A}}$ for same-state visual perturbations gives candidate-cost and top-1 stability under the preceding Lipschitz and margin assumptions; the same contraction is not desirable for state/action pairs whose external task readouts differ, which is why the discriminability guard is part of the definition rather than an auxiliary aesthetic check.

Corollary 3 (Fixed-candidate predictive stability). *Assume $D_{\text{ACPC}}^{\mathcal{A}}(h, \tilde{h}) \leq \epsilon$ for a same-state visual-perturbation pair and that J is locally L_J -Lipschitz on the rollout readout. If the clean candidate margin satisfies $\Delta > 2L_J\epsilon$, then the clean and perturbed branches select the same top-1 candidate from the fixed set $\mathcal{A}$.*

Proof. By definition of $D_{\text{ACPC}}^{\mathcal{A}}$, every candidate in $\mathcal{A}$ has rollout-readout discrepancy at most ϵ . Proposition 2 bounds the cost drift of every candidate by $L_J\epsilon$, and Proposition 3 gives top-1 stability when $\Delta > 2L_J\epsilon$. $\square$

The statement becomes selective only when it is paired with a discriminability condition such as Equation (6). Without that second condition, fixed-candidate predictive stability can still be achieved by a collapsed encoder–predictor, as shown next.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move $E_{\theta}(h)$ and $E_{\theta}(\tilde{h})$ far apart while leaving $\Pi(F_{\theta}^{1:H}(\cdot, \mathbf{a}))$ and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or cost readout has high local sensitivity.

Proposition 6 (ACPC alone permits collapse). *There exist encoder–predictor pairs for which $\text{ACPC}_H = 0$ for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical rollout readouts.*

Proof. Let $E_{\theta}(h) = z_0$ for every history and let $F_{\theta}^k(z_0, \mathbf{a}) = u_0$ for every horizon k and action sequence $\mathbf{a}$. Then clean and perturbed branches always have identical rollout readouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical readouts, violating the discriminability condition in Equation (6). $\square$

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of SMPR in the paper-facing diagnostic. This is where ACPC differs from existing nuisance-feature analyses of latent prediction: those results help explain why latent prediction may suppress irrelevant or noisy features at the representation level, whereas the condition above asks whether the remaining perturbation changes action-conditioned rollout readouts while preserving task-grounded separations.

4 Study protocol

4.1 Models and noise intervention

LeWM [19] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from $\text{Uniform}(0, \sigma_{\max})$. This is a full-sequence input-side augmentation. We sweep $\sigma_{\max} \in \{0.01, \dots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Observation+goal Gaussian noise is retained as an auxiliary stress condition.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean $\pm$ population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Table 4.

The main results follow the compressed evidence chain used in the claims. Closed-loop tables report training-seed mean/std for Gaussian robustness. ATR measures whether same-state clean/noisy rollout-disagreement tails contract at recovered endpoints, and SMPR checks whether task-grounded near-boundary separations survive that contraction. Unseen-stressor scores are auxiliary scope checks rather than additional core diagnostics.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; point-best noise-level ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 2 reports LeWM-base success rates under unperturbed and noised evaluation. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean $\pm$ population std across the three training seeds.

Table 2: LeWM-base under test-time Gaussian-noise settings. Values are mean $\pm$ population std across three training seeds; each training-seed value averages three evaluation seeds $\times$ 100 trajectories. Drop is computed per training seed as clean success minus observation-noise 0.08 success, then summarized.

Task	unpert.	obs 0.05	obs 0.08	obs+goal 0.08	drop
TwoRoom	94.67 $\pm$ 0.72	81.33 $\pm$ 6.77	68.78 $\pm$ 2.45	53.56 $\pm$ 2.57	25.89 $\pm$ 2.38
PushT	81.78 $\pm$ 4.89	19.22 $\pm$ 9.12	7.22 $\pm$ 5.81	4.89 $\pm$ 1.64	74.56 $\pm$ 5.55
Reacher	59.22 $\pm$ 1.03	33.56 $\pm$ 1.77	18.22 $\pm$ 0.42	13.78 $\pm$ 1.50	41.00 $\pm$ 1.44
Cube	65.22 $\pm$ 1.10	46.67 $\pm$ 1.96	43.11 $\pm$ 3.27	45.78 $\pm$ 0.79	22.11 $\pm$ 2.78

LeWM-base is strong on unperturbed images, but observation noise alone already produces large closed-loop drops: PushT loses 74.56 $\pm$ 5.55 pts at observation-noise 0.08, Reacher 41.00 $\pm$ 1.44 pts, TwoRoom 25.89 $\pm$ 2.38 pts, and Cube 22.11 $\pm$ 2.78 pts. The observation+goal column shows the stronger full-input-stream Gaussian-noise stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Table 3 summarizes the sweep points needed for the main argument. All entries use the unified $n = 3$ evaluation seeds (42/43/44) $\times$ 100 trajectories protocol.

Table 3: LeWM noise-sweep summary. Values are success rate mean $\pm$ population std across evaluation seeds. The two point-best columns may refer to different training-noise levels; they summarize the sweep envelope rather than a point-optimal checkpoint claim.

Task	unpert. base	obs 0.08 base	best clean row	best obs0.08 row	Main reading
TwoRoom	94.00 $\pm$ 3.56	65.67 $\pm$ 1.25	98.33 $\pm$ 0.47 (0.08)	97.67 $\pm$ 0.94 (0.08)	large recovery; visually redundant regime
PushT	86.33 $\pm$ 2.36	4.33 $\pm$ 1.25	89.67 $\pm$ 1.70 (0.03)	89.00 $\pm$ 4.32 (0.08)	large recovery; contact-sensitive state
Reacher	58.67 $\pm$ 1.25	18.33 $\pm$ 2.05	86.00 $\pm$ 2.94 (0.06)	84.67 $\pm$ 5.19 (0.07)	recovery plus fixed-epoch clean lift
Cube	66.67 $\pm$ 2.62	47.00 $\pm$ 6.38	69.33 $\pm$ 0.47 (0.01)	68.33 $\pm$ 2.49 (0.03)	weaker recovery signal

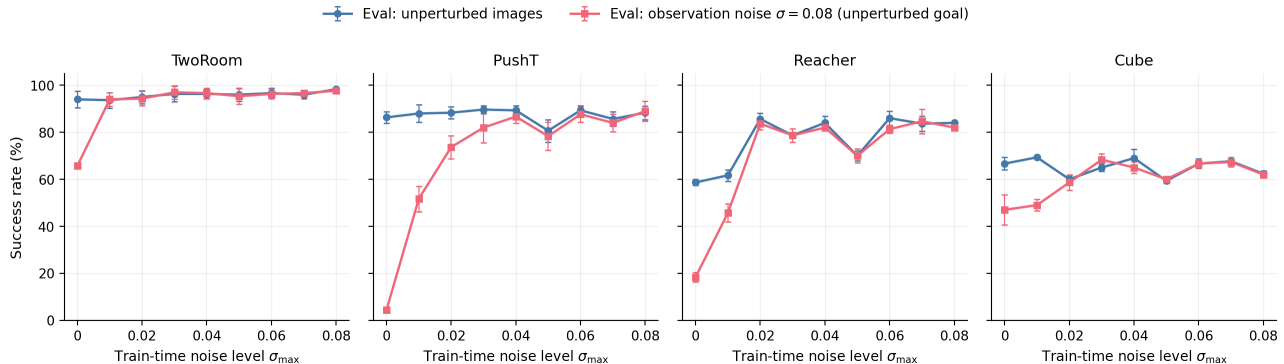


Figure 1: Noise-training sweep across four tasks. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. Error bars show the population standard deviation across the three evaluation seeds. The curves document broad task-dependent recovery plateaus rather than point-optimal ranking.

The sweep contributes two observations to the main argument. First, input-side noise can be an effective intervention in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from 18.33% to 84.67%. Second, the grid supplies representative recovered checkpoints for the downstream diagnostic question: what changed in the encoder–predictor–planner chain?

Reacher also shows a fixed-epoch clean-control lift. Noise training improves not only observation-noise robustness but also unperturbed control: the representative clean row rises from 58.67% at the no-noise baseline to about 86% under noise training. We interpret this as a fixed-protocol regularization or stabilization effect rather than pure robustness recovery. The three-training-seed replication below makes the endpoint less dependent on seed 3072, but longer learning curves would still be needed to separate regularization from optimization timing.

The completed Gaussian training-seed replication strengthens the closed-loop part of the claim. It repeats the full Gaussian sweep for LeWM seeds 3073 and 3074 and combines them with the canonical seed-3072 sweep. Table 4 summarizes the observation-only Gaussian endpoint across all three training seeds.

Table 4: Three-training-seed Gaussian replication for LeWM. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each seed point is itself the mean over evaluation seeds 42/43/44 with 100 trajectories per eval seed. The endpoint is observation-only Gaussian evaluation at 0.08 with the goal image unperturbed. Regret is best obs 0.08 minus the fixed $\sigma_{\max} = 0.08$ endpoint; the best- $\sigma_{\max}$ range shows the task/seed-dependent plateau.

Task	base obs 0.08	std0.08 obs 0.08	gain	best obs 0.08	regret	best $\sigma_{\max}$ range
TwoRoom	68.78 $\pm$ 2.46	97.11 $\pm$ 0.57	28.33 $\pm$ 2.68	97.67 $\pm$ 0.27	0.56 $\pm$ 0.79	0.05–0.08
PushT	7.22 $\pm$ 5.81	85.78 $\pm$ 3.45	78.56 $\pm$ 5.18	87.67 $\pm$ 0.98	1.89 $\pm$ 2.67	0.04–0.08
Reacher	18.22 $\pm$ 0.42	81.55 $\pm$ 0.88	63.34 $\pm$ 1.25	83.78 $\pm$ 0.63	2.22 $\pm$ 0.87	0.05–0.07
Cube	43.11 $\pm$ 3.27	62.56 $\pm$ 1.55	19.45 $\pm$ 4.53	66.44 $\pm$ 1.34	3.89 $\pm$ 2.20	0.03–0.05

Across three training seeds, the no-noise checkpoints retain large observation-noise cliffs, while the $\sigma_{\max} = 0.08$ endpoint is within 0.56–3.89 pp of each task’s best obs0.08 row. PushT remains the sharpest stress case, improving by 78.56 $\pm$ 5.18 points at the fixed high-noise endpoint; Reacher improves by 63.34 $\pm$ 1.25 points. The

nonzero regret and task-dependent best- $\sigma_{\max}$ ranges support plateau wording rather than a universal optimum at exactly $\sigma_{\max} = 0.08$.

A three-seed blur/resize score check is reported in Appendix C. It is an auxiliary severe-stressor check, not part of the matched-Gaussian claim. The next subsection asks whether the recovered Gaussian endpoints also move in the two compressed diagnostic readouts.

5.3 Compressed selective-ACPC diagnostics

For the paper-facing diagnostic, ATR is the 90th percentile of same-state clean/noisy ACPC-H rollout distance normalized by the clean transition scale. SMPR is the task-grounded near-boundary selective-margin pass rate at margin zero. Both are computed without retraining on the fixed no-noise and $\sigma_{\max} = 0.08$ checkpoints for training seeds 3072/3073/3074. Table 5 reports population mean/std across training seeds.

Table 5: Compressed selective-ACPC diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy action-conditioned rollout disagreement; lower is better. SMPR is the task-grounded near-boundary selective-margin pass rate; higher is better.

Task	ATR base	ATR std0.08	SMPR base	SMPR std0.08
TwoRoom	1.509 ± 0.010	0.111 ± 0.016	0.34 ± 0.05	0.99 ± 0.01
PushT	3.580 ± 0.260	0.247 ± 0.017	0.44 ± 0.17	1.00 ± 0.00
Reacher	2.628 ± 0.048	0.082 ± 0.005	0.73 ± 0.03	1.00 ± 0.00
Cube	2.320 ± 0.045	0.100 ± 0.007	0.45 ± 0.03	1.00 ± 0.00

The two readouts move in the intended selective direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher task-grounded margin pass rates. This does not make ATR/SMPR a point-ranking rule inside a plateau. Cube remains the useful boundary case: the compressed diagnostics improve strongly, but the closed-loop Gaussian gain and regret in Table 4 are weaker than PushT or Reacher, so closed-loop evaluation remains the behavioral authority.

SMPR uses programmatic task-state proxy labels, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and Cube uses cube-pose/goal-relation cells. Each anchor selects the closest label-crossing neighbor inside the closest 35% state-distance neighborhood and passes if that clean different-state rollout distance exceeds the same-state noisy radius. The no-noise baselines are weak under this stricter audit, while the high-noise endpoint reaches near-one pass rate; stronger hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels remain future validation.

5.4 Bounded unseen-stressor check

The main claim remains matched Gaussian-noise robustness. After freezing the Gaussian grid, we ran strongest-severity unseen-stressor score checks to test whether the fixed $\sigma_{\max} = 0.08$ endpoint transfers outside the training perturbation family. Table 6 reports behavior only. TwoRoom and Reacher improve under strongest blur, whereas PushT and Cube do not show a clear resize gain at the scale of training-seed and evaluation variance. We use this as bounded behavior outside the matched Gaussian setting, not as a universal cross-perturbation robustness claim.

Table 6: Bounded unseen-stressor score check. Values are mean $\pm$ population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from strongest-severity unseen runs. Stress Δ is the fixed $\sigma_{\max} = 0.08$ checkpoint minus the no-noise baseline.

Task	stress	baseline stress	std0.08 stress	stress Δ	reading
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38	$+43.11 \pm 8.90$	clear score gain
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10	$+49.22 \pm 3.29$	clear score gain
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38	$+2.89 \pm 17.98$	no clear gain
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57	-0.89 ± 1.40	no clear gain

6 Discussion and limitations

What the evidence supports. This is a controlled diagnostic study, not a method paper. Closed-loop evaluation establishes the behavior; ATR checks whether same-state noisy views have small action-conditioned predictive tails; SMPR checks whether task-grounded near-boundary distinctions survive that contraction. The primary statistic is the full three-training-seed Gaussian replication, with evaluation-seed variance kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims.

Scope. Gaussian pixel noise remains the training/evaluation axis. The blur/resize rows are bounded behavior outside the matched Gaussian setting, not a transfer theorem. ATR/SMPR are not a point-best $\sigma_{\max}$ predictor and do not replace closed-loop evaluation inside a broad plateau. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

What remains. The results argue against a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. Stronger discriminability evidence would require hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels, especially for PushT and Cube. Future methods can turn ATR/SMPR into objectives, but this paper keeps that as a separate method-paper path after diagnostic validity is established.

7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a selective diagnostic problem. Closed-loop behavior establishes the phenomenon; ATR measures same-state predictive-stability tails; SMPR checks task-grounded anti-collapse margins. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian robustness plateaus across three training seeds, and the two compressed diagnostics move in the expected direction at recovered endpoints.

The theory is deliberately diagnostic: sampled-pool ACPC exposes why tail risk matters and why clean margins prevent closed-loop guarantees, while local Gaussian linearization connects the measured tail to action-conditioned encoder–predictor sensitivity. The broader implication is not solved generalization; it is a concrete audit target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

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A Proofs and Calibration for ACPC Diagnostics

Reading. These proofs calibrate ATR and SMPR as diagnostics; they do not provide closed-loop control guarantees.

Proof of Proposition 2. For any fixed action sequence $\mathbf{a}$, the assumption gives $d_H(U_h(\mathbf{a}), U_{\tilde{h}}(\mathbf{a})) \leq \epsilon$. The cost readout J is L_J -Lipschitz on the rollout-readout sequence, so $|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon$. Taking the maximum over the fixed candidate set gives the claim. $\square$

Proof of Proposition 3. Let $\mathbf{a}^*$ be the clean best candidate and $\mathbf{a}^{(2)}$ the clean runner-up. If every candidate cost moves by at most $L_J\epsilon$ and the clean margin is larger than $2L_J\epsilon$, then the perturbed cost of $\mathbf{a}^*$ remains below the perturbed cost of every other candidate. Thus the top candidate is unchanged. $\square$

Proof of Theorem 1. For a sampled pool of size K , the probability that at least one candidate has clean/noisy rollout discrepancy above ϵ is at most $K\delta$ by the union bound. On the complement of that event, every candidate cost moves by at most $L_J\epsilon$. A flip can then occur only if the clean top margin is at most $2L_J\epsilon$, which contributes the margin-tail term. This proves the stated sampled-pool instability bound. $\square$

Finite-sample calibration. For a bounded Bernoulli diagnostic with empirical pass rate $\hat{p}$ over n fixed pairs, Hoeffding gives

$$\Pr \left[p < \hat{p} - \sqrt{\frac{\log(1/\alpha)}{2n}} \right] \leq \alpha. \quad (9)$$

We use this only as calibration for the fixed SMPR pair construction; nearby dataset windows can be dependent.

Local Gaussian sensitivity. For small Gaussian pixel noise ξ , a first-order expansion gives $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Experimental Details

Reading. These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \dots, 0.08\}$.

ATR is computed at each task–training-seed–checkpoint row as the 90th percentile same-state ACPC-H distance normalized by the clean transition scale. SMPR uses task-grounded near-boundary proxy labels at margin zero. The paper reports population mean/std across training seeds 3072/3073/3074.

C Unseen-Stressor Score Check

Reading. This appendix reports behavior outside the matched Gaussian family without expanding the paper’s main claim.

After the Gaussian grid and training-seed replication were frozen, we ran strongest-severity unseen-stressor score checks on all three LeWM training seeds, comparing no-noise training to $\sigma_{\max} = 0.08$ under Gaussian blur with kernel size 15 and resize/down-upsample with factor 0.25. This appendix reports behavior only; it does not broaden the main claim into cross-perturbation robustness.

Table 7: Three-training-seed unseen-stressor score check. Values are mean $\pm$ population std across LeWM training seeds 3072/3073/3074. Stress Δ is $\sigma_{\max} = 0.08$ stress success minus no-noise stress success.

Task	unseen stress	baseline stress	std0.08 stress	stress Δ	drop impr.	reading
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38	$+43.11 \pm 8.90$	$+40.89 \pm 8.23$	clear score gain
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10	$+49.22 \pm 3.29$	$+30.22 \pm 4.16$	clear score gain
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38	$+2.89 \pm 17.98$	-3.78 ± 15.56	no clear gain
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57	-0.89 ± 1.40	$+2.78 \pm 3.40$	no clear gain